

Part-of-speech tagging:

This is a classical problem in natural language processing, and is often mentioned as a “solved problem,” at least for languages with sufficient training data. A wide range of methods have been used, like HMM-based part-of-speech taggers. There are many other methods that have been successfully used: Constraint Grammar (Karlsson, 1990), Brill’s rule learning method (Brill, 1992), perceptrons (Östling, 2018), and deep neural networks. Use the Universal Dependencies treebanks version 2.15 for training and evaluation. If your tagger is fast enough, you should be able to evaluate its performance on all UD languages.